Shifted Yangians and Finite W-Algebras in Type A

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1 Introduction

Finite W-algebras may be described abstractly through quantum Hamiltonian reduction, but in type A they admit a very concrete presentation in terms of good gradings, pyramids, and shifted Yangians. These notes follow the type A approach of Brundan and Kleshchev: first one realizes the finite W-algebra as a subalgebra of $U(\mathfrak{p})$, then one uses the combinatorics of a pyramid to compare that algebra with a shifted Yangian.

The goal is to keep the construction explicit. We start with good gradings and the twisted $\mathfrak{m}$-invariants defining the finite W-algebra. We then introduce pyramids, explain how they encode the nilpotent element and the grading, and use them to define the subalgebras $\mathfrak{p}$ and $\mathfrak{m}$. Later sections will introduce shift matrices, Yangians, shifted Yangians, and the Brundan–Kleshchev isomorphism.

2 Finite W-algebras from good gradings

We begin with the construction of finite W-algebras in the form that will be used throughout these notes. Let $\mathfrak{g} = \mathfrak{gl}_N$, and let $e \in \mathfrak{g}$ be a nilpotent element. By the Jacobson–Morozov theorem, e can be included in an $\mathfrak{sl}_2$-triple $\{e, f, h\}$. The adjoint action of h then produces a $\mathbb{Z}$-grading

$$\mathfrak{g} = \bigoplus_{j \in \mathbb{Z}} \mathfrak{g}_j, \qquad \mathfrak{g}_j = \{x \in \mathfrak{g} \mid [h, x] = jx\}.$$

This grading is the motivating example, but the finite W-algebra construction is not restricted to the Jacobson–Morozov grading. The more flexible notion is that of a **good grading**.

A $\mathbb{Z}$-grading $\mathfrak{g} = \bigoplus_{j \in \mathbb{Z}} \mathfrak{g}_j$ is called a **good grading** for e if $e \in \mathfrak{g}_2$ and the map $\operatorname{ad} e : \mathfrak{g}_j \to \mathfrak{g}_{j+2}$ is injective for $j \leq -1$ and surjective for $j \geq -1$.

For a general good grading, the degree -1 component cannot simply be ignored. Since $\operatorname{ad} e : \mathfrak{g}_{-1} \to \mathfrak{g}_1$ is an isomorphism, the bilinear form

$$\langle x, y \rangle = ([x, y], e), \qquad x, y \in \mathfrak{g}_{-1},$$

is nondegenerate and skew-symmetric on $\mathfrak{g}_{-1}$. Thus $\mathfrak{g}_{-1}$ is a symplectic vector space. One then chooses a **Lagrangian subspace** $\mathfrak{l}_{\mathrm{Lag}} \subseteq \mathfrak{g}_{-1}$ and defines

$$\mathfrak{m} = \mathfrak{l}_{\mathrm{Lag}} \oplus \bigoplus_{j \leq -2} \mathfrak{g}_j.$$

This is the nilpotent Lie subalgebra used in the general construction of the finite W-algebra.

From now on, we work with an even good grading in type A. In an **even good grading**, one has $\mathfrak{g}_j = 0$ whenever j is odd. In particular, $\mathfrak{g}_{-1} = 0$, so the Lagrangian choice disappears and the preceding definition reduces to

$$\mathfrak{m} = \bigoplus_{j \leq -2} \mathfrak{g}_j, \qquad \mathfrak{p} = \bigoplus_{j \geq 0} \mathfrak{g}_j.$$

Then $\mathfrak{m}$ and $\mathfrak{p}$ are Lie subalgebras of $\mathfrak{g}$, and $\mathfrak{g} = \mathfrak{p} \oplus \mathfrak{m}$ as a vector space.

In type A, Elashvili and Kac classified good gradings in terms of pyramids. Brundan and Kleshchev use this classification to explain why, for the algebras considered here, it is enough to work with even good gradings. More precisely, every good grading in type A can be adjusted to an even good grading in such a way that the corresponding subalgebra $\mathfrak{m}$ agrees with the one obtained from the original grading for a suitable choice of Lagrangian subspace. Thus the resulting finite W-algebra is not changed by passing to the even representative.

There is one convention to keep in mind. In the abstract definition of a good grading, the nilpotent element e lies in degree 2. Later, when we describe the construction using pyramids, we will use the divided grading, where the corresponding element e_π lies in degree 1. This only changes the indexing of the grading; it does not change the finite W-algebra.

2.1 The character χ, the shifted copy $\mathfrak{m}_\chi$, and the ideal J_χ

Let $(\cdot,\cdot)$ be the trace form on $\mathfrak{g} = \mathfrak{gl}_N$. For the even good grading fixed above, define

$$\chi : \mathfrak{m} \longrightarrow \mathbb{C}, \qquad \chi(x) = (x, e).$$

This is a Lie algebra character. Indeed, homogeneous elements of $\mathfrak{m}$ have degrees at most -2, so

$$[\mathfrak{m}, \mathfrak{m}] \subseteq \bigoplus_{j \le -4} \mathfrak{g}_j.$$

The trace form pairs $\mathfrak{g}_a$ with $\mathfrak{g}_b$ trivially unless $a+b=0$. Since $e \in \mathfrak{g}_2$, it follows that $\chi([x,y]) = 0$ for $x, y \in \mathfrak{m}$. For a general good grading, the same conclusion uses the isotropy of the chosen Lagrangian in $\mathfrak{g}_{-1}$; the realization inside $U(\mathfrak{p})$ below uses the even case.

The functional χ lets us form a shifted copy of $\mathfrak{m}$ inside $U(\mathfrak{g})$:

$$\mathfrak{m}_\chi = \{x - \chi(x) \mid x \in \mathfrak{m}\} \subset U(\mathfrak{g}).$$

The map $\mathfrak{m} \to \mathfrak{m}_\chi$, $x \mapsto x - \chi(x)$, is a Lie algebra isomorphism. Indeed, if $x, y \in \mathfrak{m}$, then $\chi([x,y]) = 0$, and hence

$$[x - \chi(x), y - \chi(y)] = [x, y] = [x, y] - \chi([x, y]).$$

Thus the commutator of two shifted elements is again a shifted element.

Let J_χ be the left ideal of $U(\mathfrak{g})$ generated by $\mathfrak{m}_\chi$, so $J_\chi = U(\mathfrak{g})\mathfrak{m}_\chi$. Since $\mathfrak{g} = \mathfrak{p} \oplus \mathfrak{m}$, the PBW theorem gives a vector-space decomposition

$$U(\mathfrak{g}) = U(\mathfrak{p}) \oplus J_\chi.$$

Therefore every element $z \in U(\mathfrak{g})$ can be written uniquely as $z = a + b$, with $a \in U(\mathfrak{p})$ and $b \in J_\chi$. This defines the PBW projection $\mathrm{pr}_\chi : U(\mathfrak{g}) \to U(\mathfrak{p})$ by $\mathrm{pr}_\chi(z) = a$.

2.2 Twisted invariants

The Lie algebra $\mathfrak{m}$ acts on $U(\mathfrak{g})$ by the adjoint action. The ideal J_χ is stable under this action. To check this, it is enough to consider the generators $y - \chi(y)$ of J_χ, with $y \in \mathfrak{m}$. For $x, y \in \mathfrak{m}$,

$$[x, y - \chi(y)] = [x, y] = [x, y] - \chi([x, y]) \in \mathfrak{m}_\chi.$$

Hence the adjoint action descends to the quotient $U(\mathfrak{g})/J_\chi$.

Using the PBW identification $U(\mathfrak{g})/J_\chi \simeq U(\mathfrak{p})$, this descended action becomes a twisted action of $\mathfrak{m}$ on $U(\mathfrak{p})$:

$$x \cdot y = \mathrm{pr}_\chi([x, y]), \qquad x \in \mathfrak{m}, \quad y \in U(\mathfrak{p}).$$

The projection is essential here. The commutator $[x, y]$ lies in $U(\mathfrak{g})$, not necessarily in $U(\mathfrak{p})$, so

we project back to $U(\mathfrak{p})$.

We define the space of **twisted $\mathfrak{m}$-invariants** by

$$U(\mathfrak{p})^{\mathfrak{m},\chi} = \{y \in U(\mathfrak{p}) \mid \mathrm{pr}_\chi([x,y]) = 0 \text{ for all } x \in \mathfrak{m}\}.$$

Since $U(\mathfrak{g}) = U(\mathfrak{p}) \oplus J_\chi$, this condition is equivalent to

$$[x,y] \in J_\chi \qquad \text{for all } x \in \mathfrak{m}.$$

The next point is important because it explains why this construction produces an algebra inside $U(\mathfrak{p})$, rather than merely a vector space. Suppose $y \in U(\mathfrak{p})^{\mathfrak{m},\chi}$. Then

$$J_\chi y \subseteq J_\chi.$$

It is enough to check this on the generators $x - \chi(x)$ of the left ideal J_χ. For $x \in \mathfrak{m}$, we have

$$(x - \chi(x))y = y(x - \chi(x)) + [x,y].$$

The first term lies in J_χ because it is a left multiple of $x - \chi(x)$, while the second term lies in J_χ because y is a twisted invariant. Hence $(x - \chi(x))y \in J_\chi$, and therefore $J_\chi y \subseteq J_\chi$.

Now let $y, z \in U(\mathfrak{p})^{\mathfrak{m},\chi}$. For $x \in \mathfrak{m}$,

$$[x, yz] = [x,y]z + y[x,z].$$

Since y and z are twisted invariants, both $[x,y]$ and $[x,z]$ lie in J_χ. The first term $[x,y]z$ lies in J_χ because $J_\chi z \subseteq J_\chi$, while the second term $y[x,z]$ lies in J_χ because J_χ is a left ideal. Therefore $[x,yz] \in J_\chi$ for every $x \in \mathfrak{m}$, and so yz is again a twisted invariant.

Thus $U(\mathfrak{p})^{\mathfrak{m},\chi}$ is a subalgebra of $U(\mathfrak{p})$.

2.3 The finite W-algebra

In the even good grading setting, this is the concrete realization already used by Kostant in the regular nilpotent case and by Lynch for even nilpotent elements. In this realization, the finite W-algebra appears not only as an endomorphism algebra or a quotient construction, but as an actual subalgebra of $U(\mathfrak{p})$:

$$W(\mathfrak{g}, e) := U(\mathfrak{p})^{\mathfrak{m},\chi}.$$

In other words, $W(\mathfrak{g}, e)$ is the subalgebra of $U(\mathfrak{p})$ formed by the elements invariant under the twisted $\mathfrak{m}$-action.

This explains why the subalgebra property above is not merely a technicality. The finite W-algebra is not only defined abstractly through a quotient construction; in the PBW realization it appears concretely as a subalgebra of $U(\mathfrak{p})$. This is the form of the construction that is most convenient for type A, because the relevant even good gradings are encoded by pyramids.

Starting from a pyramid π, we will construct a nilpotent element e_π, the corresponding grading of $\mathfrak{gl}_N$, the subalgebras $\mathfrak{p}$ and $\mathfrak{m}$, and finally the finite W-algebra

$$W(\pi) = W(\mathfrak{gl}_N, e_\pi) = U(\mathfrak{p})^{\mathfrak{m},\chi}.$$

3 Pyramids and the finite W-algebra $W(\pi)$

In type A, even good gradings are described very concretely by pyramids. In this section we explain how a pyramid determines the nilpotent element e_π, the grading on $\mathfrak{g} = \mathfrak{gl}_N$, and the corresponding subalgebras $\mathfrak{p}$ and $\mathfrak{m}$.

The main point is that, once the boxes are numbered by columns, the grading becomes especially easy to read in block form. That is why the column heights, not the row lengths, are used as the block sizes for the matrices below.

3.1 Pyramids, row lengths, and column heights

A **pyramid** is a diagram of boxes arranged in rows. The rows are contiguous horizontal strips, and the left and right endpoints move inward as one goes up the pyramid. The row lengths determine the Jordan type of the nilpotent element, while the column heights determine the block decomposition that is adapted to the grading.

Let N denote the number of boxes in the diagram. Any bijection from the set of boxes of π to $\{1, \dots, N\}$ can be used to identify the boxes with the standard basis vectors of $\mathbb{C}^N$, and hence to write the corresponding matrices in $\mathfrak{gl}_N$. If one changes the numbering, the resulting matrices are conjugate by a permutation matrix. Thus the construction does not depend essentially on the numbering. However, some numberings are more convenient than others. In these notes we use the **standard numbering**, where the boxes are numbered by columns from left to right and, within each column, from top to bottom.

For a box labeled i, we write $\mathrm{row}(i)$ for its row number and $\mathrm{col}(i)$ for its column number. Rows are counted from top to bottom, while columns are counted from left to right. We use this row convention throughout, including in the shift matrix and the formulas for invariant elements.

We will use the following running example throughout the next sections. Its row lengths from top to bottom are $(1, 3, 4, 6)$, so its Jordan type is $(6, 4, 3, 1)$. Its column heights are $(1, 3, 3, 4, 2, 1)$.



Standard numbering Dynkin weights of h

Figure 1: The running pyramid. On the left is the standard numbering by columns. On the right are the weights of the Dynkin element h placed in the same boxes; the column grading is defined by the separate element h_π.

From top to bottom, the four rows contain the following box labels:

$$(8), \qquad (2,5,9), \qquad (3,6,10,12), \qquad (1,4,7,11,13,14).$$

For instance, $\mathrm{row}(7) = 4$, $\mathrm{col}(7) = 3$, $\mathrm{row}(5) = 2$, $\mathrm{col}(5) = 3$, $\mathrm{row}(8) = 1$, and $\mathrm{col}(8) = 4$.

Since the numbering is by columns, the column index sets are

$$I_1 = \{1\}, \quad I_2 = \{2,3,4\}, \quad I_3 = \{5,6,7\}, \quad I_4 = \{8,9,10,11\}, \quad I_5 = \{12,13\}, \quad I_6 = \{14\}.$$

These sets have sizes $1,3,3,4,2,1$, exactly the column heights of the pyramid.

3.2 The nilpotent element e_π

Following Brundan and Kleshchev, we define e_π by summing the matrix units corresponding to pairs of boxes that are adjacent in the same row, with the left box pointing to the right box.

In the running example, the adjacent pairs are

$$1 \to 4, \quad 4 \to 7, \quad 7 \to 11, \quad 11 \to 13, \quad 13 \to 14,$$

$$3 \to 6, \quad 6 \to 10, \quad 10 \to 12, \quad 2 \to 5, \quad 5 \to 9.$$

Therefore

$$e_\pi = e_{1,4} + e_{4,7} + e_{7,11} + e_{11,13} + e_{13,14} + e_{3,6} + e_{6,10} + e_{10,12} + e_{2,5} + e_{5,9}.$$

This is the nilpotent element attached to the pyramid π. Each row is one Jordan chain, so the Jordan type is $(6,4,3,1)$.

3.3 The grading attached to a pyramid

Two semisimple elements associated with the pyramid serve different purposes. First, the element h of an $\mathfrak{sl}_2$-triple containing e_π is obtained by placing the weights $p+1-2k$ along each row of length p, where k counts boxes from the left. These are the blue weights in Figure 1. In the running example,

$$h = \operatorname{diag}(5,2,3,3,0,1,1,0,-2,-1,-1,-3,-3,-5).$$

Each summand e_{ij} of e_π joins adjacent boxes in a row, so $h_i - h_j = 2$ and $[h, e_\pi] = 2e_\pi$. This h defines the **Dynkin grading**.

The good grading determined by the columns of π is defined by

$$h_\pi = -2\operatorname{diag}\big(\operatorname{col}(1), \dots, \operatorname{col}(N)\big).$$

Adding a scalar matrix to h_π leaves its adjoint action unchanged. We have

$$[h_\pi, e_{ij}] = 2\big(\operatorname{col}(j) - \operatorname{col}(i)\big)e_{ij}.$$

Consequently, all degrees for this grading are even. Dividing them by 2 gives the **divided grading**

$$\deg e_{ij} = \operatorname{col}(j) - \operatorname{col}(i),$$

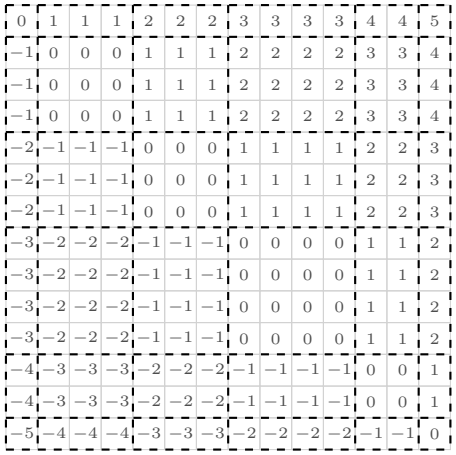
in which e_π has degree 1. The pyramid conditions ensure that the doubled grading is good for e_π; see [3, Section 4].

The elements h and h_π need not induce the same grading. For example, boxes 2 and 3 lie in the same column, but

$$[h, e_{23}] = (2-3)e_{23} = -e_{23}, \qquad [h_\pi, e_{23}] = 0.$$

Thus the Dynkin grading in this example has an odd component, while the doubled column grading is even. The definitions of $\mathfrak{p}$ and $\mathfrak{m}$ below use the column grading.

If $i \in I_r$ and $j \in I_s$, then $\deg e_{ij} = s - r$. Hence each matrix block indexed by a pair of columns has constant degree.



block sizes: 1|3|3|4|2|1

Figure 2: The full 14×14 matrix of degrees $\deg e_{ij} = \mathrm{col}(j) - \mathrm{col}(i)$ for the running example. The dashed lines show the block decomposition determined by the column heights $(1, 3, 3, 4, 2, 1)$.

Figure 2 is the visual reason for using the column heights as block sizes. Once the matrix is partitioned into blocks of sizes $1, 3, 3, 4, 2, 1$, the degree is constant on each block, and the diagonals of constant degree become immediately visible.

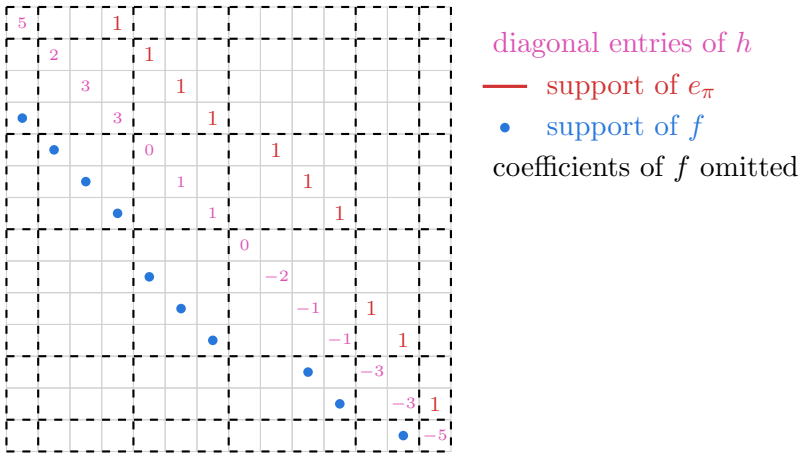


Figure 3: The same 14×14 block decomposition, now showing the diagonal entries of h, the support of e_π in red, and the support of f in blue. The dashed lines are the same column block boundaries as in Figure 2.

3.4 The subalgebras $\mathfrak{p}$, $\mathfrak{m}$, and the algebra $W(\pi)$

We now define $\mathfrak{g}_d = \mathrm{span}\{e_{ij} \mid \deg e_{ij} = d\}$ and set

$$\mathfrak{p} = \bigoplus_{d \geq 0} \mathfrak{g}_d, \qquad \mathfrak{m} = \bigoplus_{d < 0} \mathfrak{g}_d, \qquad \mathfrak{l} = \mathfrak{g}_0.$$

Thus $\mathfrak{l}$ is the Levi subalgebra, $\mathfrak{p}$ is the parabolic subalgebra, and $\mathfrak{m}$ is the nilpotent subalgebra complementary to $\mathfrak{p}$ in the grading picture.

The previous block description makes this completely transparent: $\mathfrak{p}$ consists of the block upper triangular part, $\mathfrak{m}$ consists of the block strictly lower triangular part, and $\mathfrak{l}$ is the block diagonal part.

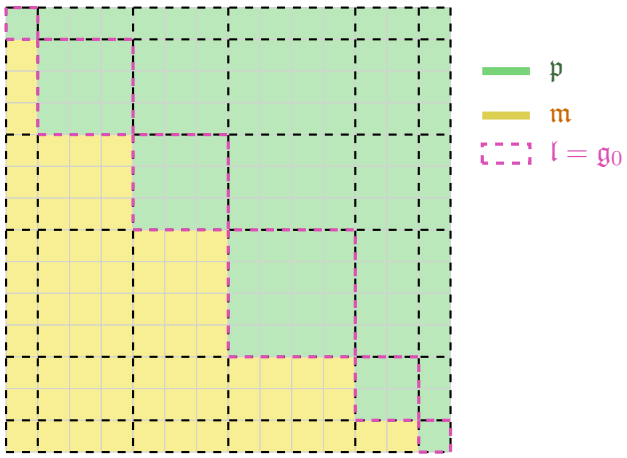


Figure 4: The parabolic subalgebra $\mathfrak{p}$ and the nilpotent subalgebra $\mathfrak{m}$ in the block decomposition determined by the pyramid. Green indicates $\mathfrak{p}$, yellow indicates $\mathfrak{m}$, and the pink dashed blocks form the Levi subalgebra $\mathfrak{l} = \mathfrak{g}_0$.

We now specialize the general construction from the previous section to the pyramid π. Using the trace form, define $\chi : \mathfrak{m} \to \mathbb{C}$ by $\chi(x) = (x, e_\pi)$. Then the corresponding finite W-algebra is

$$W(\pi) = U(\mathfrak{p})^{\mathfrak{m},\chi}.$$

This realization is particularly useful because it makes $W(\pi)$ a concrete subalgebra of $U(\mathfrak{p})$. That is exactly why it was important earlier to prove that the twisted $\mathfrak{m}$-invariants form a subalgebra of $U(\mathfrak{p})$.

From now on, whenever we speak of the finite W-algebra attached to the pyramid π, we mean the algebra $W(\pi) = U(\mathfrak{p})^{\mathfrak{m},\chi}$ constructed from the grading described above.

3.5 Shift matrices

The pyramid also determines a matrix of nonnegative integers called its shift matrix. This matrix records how the rows of the pyramid are horizontally displaced from one another.

Let π be a pyramid of level ℓ and height n, with column heights

$$(q_1, \ldots, q_\ell).$$

Choose k such that

$$0 < q_1 \leq \cdots \leq q_k, \qquad q_k \geq q_{k+1} \geq \cdots \geq q_\ell > 0.$$

The shift matrix associated with π is the $n \times n$ matrix

$$\sigma = (s_{ij})_{1 \leq i,j \leq n}$$

defined by

$$s_{ij} = \begin{cases} \#\{c = 1, \ldots, k \mid i > n - q_c \geq j\}, & i \geq j, \\ \#\{c = k+1, \ldots, \ell \mid i \leq n - q_c < j\}, & i \leq j. \end{cases}$$

There is an equivalent geometric way to read the same matrix. As throughout these notes, the rows are counted from top to bottom. If the i-th row has length p_i, let

$$L_i = \text{the number of empty columns to the left of row } i,$$

and

$$R_i = \text{the number of empty columns to the right of row } i.$$

Then

$$p_i = \ell - L_i - R_i.$$

The relation with the shift matrix is

$$L_i = s_{ni}, \qquad R_i = s_{in}.$$

More generally, the entries of the shift matrix can be read as differences of indents:

$$s_{ij} = \begin{cases} R_i - R_j, & i \leq j, \\ L_j - L_i, & i \geq j. \end{cases}$$

The shift matrix satisfies the additivity condition

$$s_{ij} + s_{jk} = s_{ik}$$

whenever

$$|i - j| + |j - k| = |i - k|.$$

Equivalently, this condition holds whenever j lies between i and k.

This follows immediately from the indent description. If $i \leq j \leq k$, then

$$s_{ij} + s_{jk} = (R_i - R_j) + (R_j - R_k) = R_i - R_k = s_{ik}.$$

Similarly, if $i \geq j \geq k$, then

$$s_{ij} + s_{jk} = (L_j - L_i) + (L_k - L_j) = L_k - L_i = s_{ik}.$$

Example 3.1. *We continue with the running pyramid whose row lengths are*

$$(6, 4, 3, 1)$$

from bottom to top, and whose column heights are

$$(q_1, q_2, q_3, q_4, q_5, q_6) = (1, 3, 3, 4, 2, 1).$$

Thus $\ell = 6$ *and* $n = 4$.

In the fixed top-to-bottom row convention, the row lengths are

$$(p_1, p_2, p_3, p_4) = (1, 3, 4, 6).$$

The following figure places the pyramid inside the ambient 4×6 *rectangle and records the left and right indents of each row.*

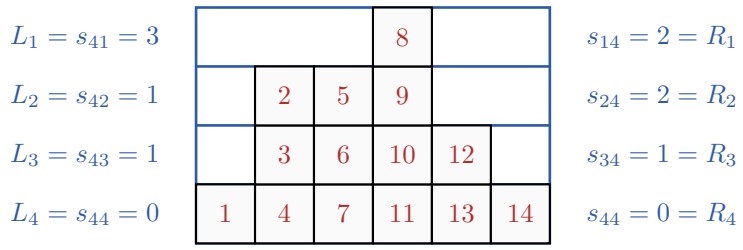


Figure 5: The shift data for the running pyramid. The ambient rectangle has height $n = 4$ and level $\ell = 6$. The left indents are $L_i = s_{4i}$, and the right indents are $R_i = s_{i4}$.

From Figure 5, the left indents are

$$L_1 = 3, \qquad L_2 = 1, \qquad L_3 = 1, \qquad L_4 = 0,$$

and the right indents are

$$R_1 = 2, \qquad R_2 = 2, \qquad R_3 = 1, \qquad R_4 = 0.$$

Equivalently,

$$L_i = s_{4i}, \qquad R_i = s_{i4}.$$

The row lengths are recovered from

$$p_i = \ell - L_i - R_i.$$

Indeed,

$$p_1 = 6-3-2 = 1, \qquad p_2 = 6-1-2 = 3, \qquad p_3 = 6-1-1 = 4, \qquad p_4 = 6-0-0 = 6.$$

Now we compute the remaining entries using the indent differences. Above the diagonal, we use the right indents:

$$s_{ij} = R_i - R_j \qquad (i \le j).$$

This gives

$$s_{12} = R_1 - R_2 = 0, \qquad s_{13} = R_1 - R_3 = 1, \qquad s_{14} = R_1 - R_4 = 2,$$

$$s_{23} = R_2 - R_3 = 1, \qquad s_{24} = R_2 - R_4 = 2, \qquad s_{34} = R_3 - R_4 = 1.$$

Below the diagonal, we use the left indents:

$$s_{ij} = L_j - L_i \qquad (i \ge j).$$

This gives

$$s_{21} = L_1 - L_2 = 2, \qquad s_{31} = L_1 - L_3 = 2, \qquad s_{41} = L_1 - L_4 = 3,$$

$$s_{32} = L_2 - L_3 = 0, \qquad s_{42} = L_2 - L_4 = 1, \qquad s_{43} = L_3 - L_4 = 1.$$

Therefore the shift matrix is

$$\sigma = \begin{pmatrix} 0 & 0 & 1 & 2 \\ 2 & 0 & 1 & 2 \\ 2 & 0 & 0 & 1 \\ 3 & 1 & 1 & 0 \end{pmatrix}.$$

4 Yangians

We now recall the ordinary Yangian associated with $\mathfrak{gl}_n$. The purpose of this section is to introduce the algebra $Y_n = Y(\mathfrak{gl}_n)$, first through the RTT presentation and then through the Gauss decomposition of the matrix $T(u)$. The shifted Yangian will be obtained later by modifying this ordinary picture using the shift matrix σ.

4.1 The RTT presentation

Let E_{ij} denote the elementary matrix units in $\mathrm{End}(\mathbb{C}^n)$, and let

$$P = \sum_{i,j=1}^{n} E_{ij} \otimes E_{ji}$$

be the permutation operator on $\mathbb{C}^n \otimes \mathbb{C}^n$. We use the rational R-matrix

$$R(u) = u - P.$$

This matrix satisfies the Yang-Baxter equation, and it is the input for the RTT presentation of the Yangian.

Define formal power series

$$T_{ij}(u) = \delta_{ij} + \sum_{r>0} T_{ij}^{(r)} u^{-r}, \qquad 1 \le i, j \le n,$$

and assemble them into the $n \times n$ matrix

$$T(u) = \big(T_{ij}(u)\big)_{1 \le i,j \le n}.$$

Equivalently, one may view $T(u)$ as an element of

$$\mathrm{End}(\mathbb{C}^n) \otimes Y_n[[u^{-1}]]$$

by writing

$$T(u) = \sum_{i,j=1}^{n} E_{ij} \otimes T_{ij}(u).$$

The Yangian $Y_n = Y(\mathfrak{gl}_n)$ is the associative algebra over $\mathbb{C}$ generated by the elements

$$T_{ij}^{(r)}, \qquad 1 \le i, j \le n, \qquad r > 0,$$

subject to the RTT relation

$$R^{12}(u - v) T^1(u) T^2(v) = T^2(v) T^1(u) R^{12}(u - v).$$

Here

$$T^1(u) = \sum_{i,j=1}^{n} E_{ij} \otimes 1 \otimes T_{ij}(u), \qquad T^2(v) = \sum_{i,j=1}^{n} 1 \otimes E_{ij} \otimes T_{ij}(v),$$

so the RTT relation is an identity in

$$\mathrm{End}(\mathbb{C}^n) \otimes \mathrm{End}(\mathbb{C}^n) \otimes Y_n[[u^{-1}, v^{-1}]].$$

Equating coefficients gives the relations

$$[T_{ij}^{(r)}, T_{hk}^{(s)}] = \sum_{t=0}^{\min(r,s)-1} \Big(T_{ik}^{(r+s-1-t)} T_{hj}^{(t)} - T_{ik}^{(t)} T_{hj}^{(r+s-1-t)} \Big),$$

for all $1 \le h, i, j, k \le n$ and all $r, s > 0$, with the convention

$$T_{ij}^{(0)} = \delta_{ij}.$$

This coefficient form of the RTT relation is the presentation that will be used below.

4.2 Gauss decomposition and Drinfeld generators

The RTT generators are natural from the R-matrix point of view, but another presentation is better adapted to shifted Yangians and to the comparison with finite W-algebras. This presentation comes from the Gauss decomposition of $T(u)$.

We factor $T(u)$ as

$$T(u) = F(u)D(u)E(u),$$

where

$$D(u) = \begin{pmatrix} D_1(u) & & & 0 \\ & D_2(u) & & \\ & & \ddots & \\ 0 & & & D_n(u) \end{pmatrix}$$

is diagonal,

$$E(u) = \begin{pmatrix} 1 & E_{12}(u) & E_{13}(u) & \cdots & E_{1n}(u) \\ 0 & 1 & E_{23}(u) & \cdots & E_{2n}(u) \\ \vdots & & \ddots & \ddots & \vdots \\ 0 & \cdots & 0 & 1 & E_{n-1,n}(u) \\ 0 & \cdots & \cdots & 0 & 1 \end{pmatrix}$$

is upper unitriangular, and

$$F(u) = \begin{pmatrix} 1 & 0 & \cdots & \cdots & 0 \\ F_{21}(u) & 1 & 0 & \cdots & 0 \\ F_{31}(u) & F_{32}(u) & 1 & \cdots & 0 \\ \vdots & \vdots & \ddots & \ddots & 0 \\ F_{n1}(u) & F_{n2}(u) & \cdots & F_{n,n-1}(u) & 1 \end{pmatrix}$$

is lower unitriangular.

We write

$$D_i(u) = 1 + \sum_{r>0} D_i^{(r)} u^{-r}.$$

For the simple upper and lower off-diagonal entries, define

$$E_i(u) := E_{i,i+1}(u), \qquad F_i(u) := F_{i+1,i}(u),$$

and expand

$$E_i(u) = \sum_{r>0} E_i^{(r)} u^{-r}, \qquad F_i(u) = \sum_{r>0} F_i^{(r)} u^{-r}.$$

The coefficients

$$D_i^{(r)}, \qquad E_i^{(r)}, \qquad F_i^{(r)}$$

are the Drinfeld generators of the Yangian.

In this notation, the $D_i^{(r)}$'s play the role of Cartan-type generators, the $E_i^{(r)}$'s play the role of positive root generators, and the $F_i^{(r)}$'s play the role of negative root generators. The defining relations among these generators are obtained by translating the RTT relation through the Gauss decomposition.

4.3 The Drinfeld relations

To state the relations compactly, define the inverse series

$$\widetilde{D}_i(u) = -D_i(u)^{-1} = \sum_{r\geq 0} \widetilde{D}_i^{(r)} u^{-r}.$$

Thus

$$\widetilde{D}_i^{(0)} = -1.$$

The Drinfeld generators satisfy the following relations:

$$[D_i^{(r)}, D_j^{(s)}] = 0,$$

$$[E_i^{(r)}, F_j^{(s)}] = \delta_{ij} \sum_{t=0}^{r+s-1} \widetilde{D}_i^{(t)} D_{i+1}^{(r+s-1-t)},$$

$$[D_i^{(r)}, E_j^{(s)}] = (\delta_{ij} - \delta_{i,j+1}) \sum_{t=0}^{r-1} D_i^{(t)} E_j^{(r+s-1-t)},$$

and

$$[D_i^{(r)}, F_j^{(s)}] = (\delta_{i,j+1} - \delta_{ij}) \sum_{t=0}^{r-1} F_j^{(r+s-1-t)} D_i^{(t)}.$$

There are also relations among modes of the same simple-root generator. One has

$$[E_i^{(r)}, E_i^{(s+1)}] - [E_i^{(r+1)}, E_i^{(s)}] = E_i^{(r)} E_i^{(s)} + E_i^{(s)} E_i^{(r)},$$

and

$$[F_i^{(r+1)}, F_i^{(s)}] - [F_i^{(r)}, F_i^{(s+1)}] = F_i^{(r)} F_i^{(s)} + F_i^{(s)} F_i^{(r)}.$$

For neighboring simple roots, $1 \leq i \leq n-2$, the mode relations are

$$[E_i^{(r)}, E_{i+1}^{(s+1)}] - [E_i^{(r+1)}, E_{i+1}^{(s)}] = -E_i^{(r)} E_{i+1}^{(s)},$$

and

$$[F_i^{(r+1)}, F_{i+1}^{(s)}] - [F_i^{(r)}, F_{i+1}^{(s+1)}] = -F_{i+1}^{(s)} F_i^{(r)}.$$

All modes here are positive integers. For non-neighboring simple roots, the relations are

$$[E_i^{(r)}, E_j^{(s)}] = 0 \qquad \text{if } |i-j| > 1,$$

and

$$[F_i^{(r)}, F_j^{(s)}] = 0 \qquad \text{if } |i-j| > 1.$$

Finally, the Serre relations are

$$[E_i^{(r)}, [E_i^{(s)}, E_j^{(t)}]] + [E_i^{(s)}, [E_i^{(r)}, E_j^{(t)}]] = 0 \qquad \text{if } |i-j| = 1,$$

and

$$[F_i^{(r)}, [F_i^{(s)}, F_j^{(t)}]] + [F_i^{(s)}, [F_i^{(r)}, F_j^{(t)}]] = 0 \qquad \text{if } |i-j| = 1.$$

These relations give the Drinfeld presentation of the ordinary Yangian Y_n; see [2, Theorem 5.2]. With the convention $\widetilde{D}_i(u) = -D_i(u)^{-1}$, the mixed relation at $r = s = 1$ reads $[E_i^{(1)}, F_i^{(1)}] = D_i^{(1)} - D_{i+1}^{(1)}$, in agreement with $[e_{i,i+1}, e_{i+1,i}] = e_{ii} - e_{i+1,i+1}$. The RTT generators $T_{ij}^{(r)}$ and the Drinfeld generators $D_i^{(r)}, E_i^{(r)}, F_i^{(r)}$ are two ways of presenting the same algebra. The bridge between them is the Gauss decomposition

$$T(u) = F(u)D(u)E(u).$$

4.4 Parabolic presentation

We now introduce a block version of the Gauss decomposition of $T(u)$. The extra datum is a choice of block sizes.

A **shape** is a tuple

$$\nu = (\nu_1, \dots, \nu_m)$$

of positive integers such that

$$\nu_1 + \cdots + \nu_m = n.$$

Thus a shape is simply a composition of n. It determines a decomposition of the index set

$$\{1, \dots, n\}$$

into consecutive blocks

$$B_1 = \{1, \dots, \nu_1\},$$

$$B_2 = \{\nu_1 + 1, \dots, \nu_1 + \nu_2\},$$

and, in general,

$$B_a = \{\nu_1 + \cdots + \nu_{a-1} + 1, \dots, \nu_1 + \cdots + \nu_a\}.$$

Equivalently, ν is the block-size data for the standard Levi subalgebra

$$\mathfrak{gl}_{\nu_1} \oplus \cdots \oplus \mathfrak{gl}_{\nu_m} \subseteq \mathfrak{gl}_n.$$

Once a shape ν has been fixed, we regard the matrix $T(u) = \big(T_{ij}(u)\big)_{1\le i,j\le n}$ as an $m \times m$ block matrix, where the (a,b)-block has size $\nu_a \times \nu_b$.

The parabolic presentation is obtained by applying Gauss decomposition to this block matrix.

The usual Gauss decomposition considered above corresponds to the special shape

$$\nu = (1, 1, \ldots, 1).$$

In that case, every block has size 1, and the corresponding generators are the usual Drinfeld generators

$$D_i^{(r)}, \qquad E_i^{(r)}, \qquad F_i^{(r)}.$$

The parabolic presentation is obtained by grouping the rows and columns of $T(u)$ into larger consecutive blocks before performing the same Gauss decomposition.

The block Gauss decomposition has the form

$$T(u) = F(u)D(u)E(u),$$

where $D(u)$ is block diagonal, $E(u)$ is block upper unitriangular, and $F(u)$ is block lower unitriangular.

The diagonal factor $D(u)$ has diagonal blocks

$$D_1(u), D_2(u), \ldots, D_m(u),$$

where $D_a(u)$ is a $\nu_a \times \nu_a$ matrix for $1 \le a \le m$. The off-diagonal blocks of $D(u)$ are zero.

The upper unitriangular factor $E(u)$ has identity blocks on the diagonal and zero blocks below the diagonal. We denote the adjacent upper-diagonal blocks by $E_a(u), 1 \le a < m$. Here $E_a(u)$ has size $\nu_a \times \nu_{a+1}$.

The remaining upper off-diagonal blocks are determined by the product $F(u)D(u)E(u)$, so they are not taken as generators.

Similarly, the lower unitriangular factor $F(u)$ has identity blocks on the diagonal and zero blocks above the diagonal. We denote the adjacent lower-diagonal blocks by $F_a(u), 1 \le a < m$. Here $F_a(u)$ has size $\nu_{a+1} \times \nu_a$.

The remaining lower off-diagonal blocks are also determined by the product $F(u)D(u)E(u)$, so they are not taken as generators.

For $1 \le a \le m$, write

$$D_a(u) = \big(D_{a;i,j}(u)\big)_{1\le i,j\le \nu_a}.$$

For $1 \leq a < m$, write

$$E_a(u) = \big(E_{a;i,j}(u)\big)_{1\leq i\leq \nu_a,\ 1\leq j\leq \nu_{a+1}},$$

and

$$F_a(u) = \big(F_{a;i,j}(u)\big)_{1\leq i\leq \nu_{a+1},\ 1\leq j\leq \nu_a}.$$

The entries are expanded as

$$D_{a;i,j}(u) = \sum_{r\geq 0} D^{(r)}_{a;i,j} u^{-r}, \qquad 1 \leq a \leq m, \quad 1 \leq i,j \leq \nu_a,$$

$$E_{a;i,j}(u) = \sum_{r> 0} E^{(r)}_{a;i,j} u^{-r}, \qquad 1 \leq a < m, \quad 1 \leq i \leq \nu_a, \quad 1 \leq j \leq \nu_{a+1},$$

and

$$F_{a;i,j}(u) = \sum_{r> 0} F^{(r)}_{a;i,j} u^{-r}, \qquad 1 \leq a < m, \quad 1 \leq i \leq \nu_{a+1}, \quad 1 \leq j \leq \nu_a.$$

Since $D_a(u)$ has constant term I_{ν_a}, we have

$$D^{(0)}_{a;i,j} = \delta_{ij}.$$

We also introduce the inverse matrix series

$$\widetilde{D}_a(u) = -D_a(u)^{-1}.$$

Writing

$$\widetilde{D}_a(u) = \big(\widetilde{D}_{a;i,j}(u)\big)_{1\leq i,j\leq \nu_a},$$

and

$$\widetilde{D}_{a;i,j}(u) = \sum_{r\geq 0} \widetilde{D}^{(r)}_{a;i,j} u^{-r},$$

we have

$$\widetilde{D}^{(0)}_{a;i,j} = -\delta_{ij}.$$

The parabolic generators of Y_n associated with the shape ν are

$$D^{(r)}_{a;i,j} \qquad 1 \leq a \leq m, \quad 1 \leq i,j \leq \nu_a, \quad r > 0,$$

$$E^{(r)}_{a;i,j} \qquad 1 \leq a < m, \quad 1 \leq i \leq \nu_a, \quad 1 \leq j \leq \nu_{a+1}, \quad r > 0,$$

and

$$F^{(r)}_{a;i,j} \qquad 1 \leq a < m, \quad 1 \leq i \leq \nu_{a+1}, \quad 1 \leq j \leq \nu_a, \quad r > 0.$$

At the other extreme, if $\nu = (n)$, then there is only one block and $D_1(u) = T(u)$, so the parabolic presentation reduces to the RTT presentation.

Thus the parabolic presentation interpolates between the RTT presentation and the Drinfeld

presentation, but it does not define a new algebra. It is another way to present the same Yangian Y_n, obtained by applying Gauss decomposition after choosing a block decomposition of the matrix $T(u)$.

5 Shifted Yangians

We now introduce the shifted Yangian associated with a shift matrix $\sigma = (s_{ij})$. The ordinary Yangian Y_n has generators $D_i^{(r)}$, $E_i^{(r)}$, and $F_i^{(r)}$ for all positive modes r. The **shifted Yangian** is obtained by keeping all the D-generators, but removing the first few E- and F-generators according to the shifts.

Fix $n \geq 1$ and a shift matrix $\sigma = (s_{ij})_{1 \leq i,j \leq n}$. The shifted Yangian associated with σ is denoted $Y_n(\sigma)$. It is the algebra over $\mathbb{C}$ generated by

$$D_i^{(r)}, \qquad 1 \leq i \leq n, \quad r > 0,$$

together with

$$E_i^{(r)}, \qquad 1 \leq i < n, \quad r > s_{i,i+1},$$

and

$$F_i^{(r)}, \qquad 1 \leq i < n, \quad r > s_{i+1,i}.$$

The defining relations are the same Drinfeld relations as for the ordinary Yangian, but only for the generators that remain after imposing these lower bounds on the modes.

Thus the shifts affect only the starting modes of the E- and F-generators. The $D_i^{(r)}$ are never shifted. If $\sigma = 0$, then $s_{ij} = 0$ for all i, j, so the conditions above become simply $r > 0$. Therefore $Y_n(0)$ is the ordinary Yangian Y_n.

5.1 Admissible shapes

The parabolic presentation becomes useful for shifted Yangians only after choosing a shape compatible with the shift matrix.

Let $\nu = (\nu_1, \ldots, \nu_m)$ be a shape for n, and let

$$B_a = \{\nu_1 + \cdots + \nu_{a-1} + 1, \ldots, \nu_1 + \cdots + \nu_a\}$$

be the corresponding consecutive blocks. We say that ν is **admissible** for σ if $s_{ij} = 0$ whenever i and j lie in the same block B_a.

In other words, an admissible shape groups together only indices between which there is no internal shift. The shifts are then concentrated between different blocks.

The **minimal admissible shape** is obtained by taking the largest possible consecutive blocks on which all shifts vanish internally. Any refinement of an admissible shape is again admissible. In particular, the shape $(1, 1, \ldots, 1)$ is always admissible.

For an admissible shape ν, define

$$\nu_{<a} = \nu_1 + \cdots + \nu_{a-1}.$$

The shift between two blocks is then measured by

$$s_{a,b}(\nu) = s_{\nu_{<a}+\nu_a,\ \nu_{<b}+\nu_b}.$$

Because the entries of a shift matrix satisfy the additivity rule, this number is independent of which indices are chosen inside the corresponding blocks. Thus $s_{a,b}(\nu)$ records the shift from block a to block b.

Example 5.1. *Consider the pyramid displayed on the left of Figure 6. It has four rows: the bottom two rows have length 4, while the top two rows have length 2 and are centered inside the bottom rows. The red boundary indicates the indentation of the top two rows.*

The shift matrix associated with this pyramid is

$$\sigma = \begin{pmatrix} 0 & 0 & 1 & 1 \\ 0 & 0 & 1 & 1 \\ 1 & 1 & 0 & 0 \\ 1 & 1 & 0 & 0 \end{pmatrix}.$$

The zero upper-left block shows that the indices $1, 2$ may be grouped together, and the zero lower-right block shows that the indices $3, 4$ may be grouped together. Hence the minimal admissible shape is $(2, 2)$.

The shape shown in Figure 6 is

$$\nu = (2, 1, 1).$$

It is admissible because it is a refinement of $(2, 2)$. The other admissible refinements of $(2, 2)$ are

$$(1, 1, 2) \qquad \textit{and} \qquad (1, 1, 1, 1).$$

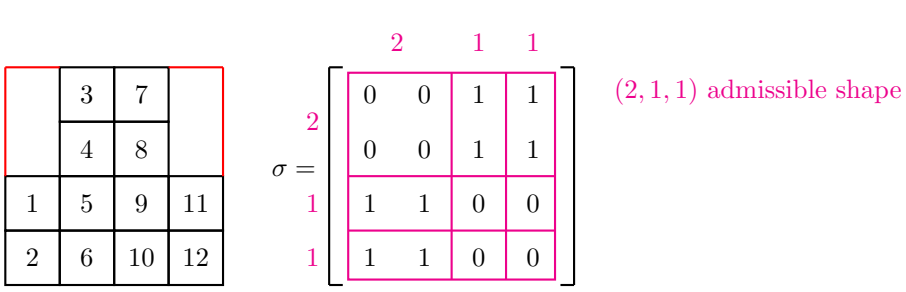


Figure 6: An admissible shape for the shift matrix σ. The magenta partition shows that $\nu = (2, 1, 1)$ is admissible.

In what follows, whenever we use a parabolic presentation for $Y_n(\sigma)$, the shape will be assumed to be admissible for σ.

5.2 Parabolic generators for $Y_n(\sigma)$

Let $\nu = (\nu_1, \dots, \nu_m)$ be an admissible shape for σ. The parabolic generators for the shifted Yangian are the same block generators as in the ordinary Yangian, but the lower bounds on the modes of E and F are shifted.

The generators are

$$D_{a;i,j}^{(r)} \qquad 1 \le a \le m, \quad 1 \le i, j \le \nu_a, \quad r > 0,$$

together with

$$E_{a;i,j}^{(r)} \qquad 1 \le a < m, \quad 1 \le i \le \nu_a, \quad 1 \le j \le \nu_{a+1}, \quad r > s_{a,a+1}(\nu),$$

and

$$F_{a;i,j}^{(r)} \qquad 1 \le a < m, \quad 1 \le i \le \nu_{a+1}, \quad 1 \le j \le \nu_a, \quad r > s_{a+1,a}(\nu).$$

Set $D_a^{(0)} = I_{\nu_a}$ and introduce the matrix series

$$D_a(u) = \sum_{r \ge 0} D_a^{(r)} u^{-r}, \qquad D_a^{(r)} = (D_{a;i,j}^{(r)}), \qquad \widetilde{D}_a(u) = -D_a(u)^{-1}.$$

The complete defining relations are the shifted parabolic relations [1, (3.3)–(3.14)]. Their internal matrix indices are essential whenever some $\nu_a > 1$. For example,

$$[D_{a;i,j}^{(r)}, D_{b;h,k}^{(s)}] = \delta_{ab} \sum_{t=0}^{\min(r,s)-1} \big(D_{a;i,k}^{(r+s-1-t)} D_{a;h,j}^{(t)} - D_{a;i,k}^{(t)} D_{a;h,j}^{(r+s-1-t)} \big),$$

$$[E_{a;i,j}^{(r)}, F_{b;h,k}^{(s)}] = \delta_{ab} \sum_{t=0}^{r+s-1} \widetilde{D}_{a;i,k}^{(t)} D_{a+1;h,j}^{(r+s-1-t)}.$$

Thus entries in the same diagonal block need not commute. When $\nu = (1, \dots, 1)$, the block relations reduce to the scalar Drinfeld relations from Section 4.3, with the shifted mode ranges. For the zero shift matrix, they give the ordinary parabolic presentation from Section 4.4. In the sum form of the same-root E, E and F, F relations, the lower summation bounds are respectively $s_{a,a+1}(\nu) + 1$ and $s_{a+1,a}(\nu) + 1$.

5.3 Embedding in the ordinary Yangian

There is a canonical algebra homomorphism

$$\iota : Y_n(\sigma) \longrightarrow Y_n$$

which sends every allowed Drinfeld generator to the element with the same name. This map is injective by the PBW theorem for shifted Yangians [1, Theorem 2.1 and Corollary 2.2]. To explain the filtration argument, give $D_i^{(r)}$, $E_i^{(r)}$, and $F_i^{(r)}$ loop degree $r-1$. The associated graded algebra is

$$\operatorname{gr}^{\mathrm{loop}} Y_n(\sigma) \cong U(\mathfrak{gl}_n[t](\sigma)), \qquad \mathfrak{gl}_n[t](\sigma) = \operatorname{span}\{e_{ij}t^r \mid r \geq s_{ij}\}.$$

The shift-matrix condition implies $s_{ij} + s_{jk} \geq s_{ik}$ for arbitrary i, j, k, with equality when j lies between i and k. Hence $\mathfrak{gl}_n[t](\sigma)$ is a Lie subalgebra of $\mathfrak{gl}_n[t]$. Under the PBW identifications, the associated graded map of ι is the inclusion

$$U(\mathfrak{gl}_n[t](\sigma)) \hookrightarrow U(\mathfrak{gl}_n[t]).$$

Injectivity of the associated graded map implies injectivity of ι. We therefore identify $Y_n(\sigma)$ with this subalgebra of Y_n. For an admissible shape, the allowed parabolic generators inside Y_n generate the same subalgebra; see [1, Theorem 3.2 and Corollary 3.3].

6 Shifted Yangians of level ℓ

We now pass from the shifted Yangian $Y_n(\sigma)$ to a finite-level quotient. This is the algebra that will later be identified with the finite W-algebra attached to a pyramid.

Fix a shift matrix $\sigma = (s_{ij})_{1 \leq i,j \leq n}$ and an integer ℓ such that $\ell \geq s_{1n} + s_{n1}$. Define

$$p_i = \ell - s_{in} - s_{ni}, \qquad 1 \leq i \leq n.$$

Then

$$0 \leq p_1 \leq p_2 \leq \cdots \leq p_n = \ell.$$

If σ comes from a pyramid of level ℓ, then p_i is the length of the i-th row, where rows are counted from top to bottom. Indeed, s_{ni} is the left indent of row i, while s_{in} is the right indent of row i, so

$$p_i = \ell - s_{ni} - s_{in}.$$

The **shifted Yangian of level** ℓ, denoted $Y_{n,\ell}(\sigma)$, is the quotient

$$Y_{n,\ell}(\sigma) = Y_n(\sigma)/I_\ell,$$

where I_ℓ is the two-sided ideal generated by

$$D_1^{(r)} \qquad \text{for } r > p_1.$$

Thus the level does not change the shifted Yangian relations themselves. It imposes an additional truncation.

Let $\nu = (\nu_1, \dots, \nu_m)$ be an admissible shape for σ. In the parabolic presentation, the same quotient may be described as the quotient of $Y_n(\sigma)$ by the two-sided ideal generated by

$$D^{(r)}_{1;i,j} \qquad 1 \le i, j \le \nu_1, \quad r > p_1.$$

This agrees with the previous definition because $D^{(r)}_{1;1,1} = D^{(r)}_1$, and the remaining entries in the first diagonal block are forced by the parabolic relations.

From now on, we use the same symbols

$$D^{(r)}_{a;i,j}, \qquad E^{(r)}_{a;i,j}, \qquad F^{(r)}_{a;i,j}$$

for the parabolic generators of $Y_n(\sigma)$ and for their images in the quotient $Y_{n,\ell}(\sigma)$. With this convention, the relations in $Y_{n,\ell}(\sigma)$ are the images of the shifted parabolic relations, together with the truncation relations

$$D^{(r)}_{1;i,j} = 0 \qquad 1 \le i, j \le \nu_1, \quad r > p_1.$$

Thus any relation involving one of these elements is also understood modulo the quotient ideal. In particular, some terms that appear in the non-truncated shifted Yangian become zero after passing to level ℓ.

For an admissible shape ν, define

$$p_a(\nu) = p_{\nu_1 + \cdots + \nu_a}, \qquad 1 \le a \le m.$$

Since ν is admissible, the numbers p_i are constant on each block B_a, so $p_a(\nu)$ is the row length attached to the a-th block.

To state the PBW basis, we also need the non-adjacent parabolic root generators. These non-adjacent elements are not additional defining generators. They are root elements constructed from the adjacent generators, and they are the elements that appear naturally in the PBW basis. For $1 \le a < b \le m$, define $E^{(r)}_{a,b;i,j}$ recursively by

$$E^{(r)}_{a,a+1;i,j} := E^{(r)}_{a;i,j}$$

and

$$E^{(r)}_{a,b;i,j} := \left[E^{(r - s_{b-1,b}(\nu))}_{a,b-1;i,k}, E^{(s_{b-1,b}(\nu)+1)}_{b-1;k,j} \right],$$

where $1 \le k \le \nu_{b-1}$. Similarly, define $F^{(r)}_{a,b;i,j}$ by

$$F^{(r)}_{a,a+1;i,j} := F^{(r)}_{a;i,j}$$

and

$$F^{(r)}_{a,b;i,j} := \left[F^{(s_{b,b-1}(\nu)+1)}_{b-1;i,k}, F^{(r - s_{b,b-1}(\nu))}_{a,b-1;k,j} \right],$$

where $1 \le k \le \nu_{b-1}$. These elements are independent of the choice of k.

The following is the PBW theorem for the level ℓ quotient.

Theorem 6.1. *Let $\nu = (\nu_1, \dots, \nu_m)$ be an admissible shape for σ. The monomials, taken in any fixed order, in the following elements form a basis of $Y_{n,\ell}(\sigma)$:*

$$D^{(r)}_{a;i,j} \qquad 1 \le a \le m, \quad 1 \le i, j \le \nu_a, \quad 0 < r \le p_a(\nu),$$

$$E^{(r)}_{a,b;i,j} \qquad 1 \le a < b \le m, \quad 1 \le i \le \nu_a, \quad 1 \le j \le \nu_b, \quad s_{a,b}(\nu) < r \le s_{a,b}(\nu) + p_a(\nu),$$

and

$$F^{(r)}_{a,b;i,j} \qquad 1 \le a < b \le m, \quad 1 \le i \le \nu_b, \quad 1 \le j \le \nu_a, \quad s_{b,a}(\nu) < r \le s_{b,a}(\nu) + p_a(\nu).$$

We will use this theorem as a quoted result from Section 6 of [1]. The important point for these notes is that $Y_{n,\ell}(\sigma)$ is controlled by a finite list of PBW root elements. Ordered monomials in those elements form a basis. The defining ideal only mentions the high modes of the first diagonal block, but the relations imply that all elements outside the ranges above are either zero in the quotient or expressible in terms of the listed generators.

There is also a useful connection with the non-truncated shifted Yangian. The quotients form an inverse system

$$\cdots \longrightarrow Y_{n,\ell+2}(\sigma) \longrightarrow Y_{n,\ell+1}(\sigma) \longrightarrow Y_{n,\ell}(\sigma).$$

Equip $Y_n(\sigma)$ with its **canonical filtration**, in which each defining generator of mode r has degree r, and give each quotient the induced filtration. This differs from the loop filtration used for the embedding above. Then

$$Y_n(\sigma) \cong \varprojlim_{\ell}^{\text{filtered}} Y_{n,\ell}(\sigma),$$

where the inverse limit is taken in the category of filtered algebras [1, Remark 6.4]. Equivalently, only compatible families of uniformly bounded canonical filtration degree are retained:

$$Y_n(\sigma) \cong \bigcup_{d \ge 0} \varprojlim_{\ell} F_d Y_{n,\ell}(\sigma).$$

The unrestricted inverse limit can be larger. For $n = 1$ and $\sigma = 0$, the family $\sum_{r=1}^{\ell} D^{(r)}$ is compatible in the level quotients, but it is not represented by a polynomial in Y_1. Its canonical filtration degrees are unbounded.

Remark 6.2. *When $\sigma = 0$, all shifts vanish and $p_i = \ell$ for every i. In this case $Y_{n,\ell}(0)$ is the ordinary Yangian of level ℓ. Equivalently, it is the quotient of Y_n by the two-sided ideal generated by*

$$T^{(r)}_{ij} \qquad 1 \le i, j \le n, \quad r > \ell.$$

This is the rectangular case, where the corresponding pyramid has n rows all of length ℓ.

Example 6.3. *Return to the pyramid from Figure 6. Its level is $\ell = 4$, and its row lengths from top to bottom are*

$$(p_1, p_2, p_3, p_4) = (2, 2, 4, 4).$$

The associated shift matrix is

$$\sigma = \begin{pmatrix} 0 & 0 & 1 & 1 \\ 0 & 0 & 1 & 1 \\ 1 & 1 & 0 & 0 \\ 1 & 1 & 0 & 0 \end{pmatrix}.$$

We use the admissible shape

$$\nu = (2, 1, 1).$$

Thus the blocks are

$$B_1 = \{1, 2\}, \qquad B_2 = \{3\}, \qquad B_3 = \{4\}.$$

For this shape,

$$p_1(\nu) = 2, \qquad p_2(\nu) = 4, \qquad p_3(\nu) = 4.$$

The block shifts are

$$s_{1,2}(\nu) = 1, \qquad s_{1,3}(\nu) = 1, \qquad s_{2,3}(\nu) = 0,$$

and

$$s_{2,1}(\nu) = 1, \qquad s_{3,1}(\nu) = 1, \qquad s_{3,2}(\nu) = 0.$$

The shifted Yangian of level 4 attached to this pyramid is $Y_{4,4}(\sigma)$. In the parabolic presentation associated with $\nu = (2, 1, 1)$, it is the quotient of $Y_4(\sigma)$ by the two-sided ideal generated by

$$D^{(r)}_{1;i,j} \qquad 1 \le i, j \le 2, \quad r > 2.$$

As above, we use the same symbols for the images of the parabolic generators in the quotient.

The PBW theorem says that ordered monomials in the following elements form a basis of $Y_{4,4}(\sigma)$.

For the diagonal part, the elements are

$$D^{(r)}_{1;i,j} \qquad 1 \le i, j \le 2, \quad 1 \le r \le 2,$$

together with

$$D^{(r)}_{2;1,1} \qquad 1 \le r \le 4,$$

and

$$D^{(r)}_{3;1,1} \qquad 1 \le r \le 4.$$

For the positive root elements, the allowed elements are

$$E^{(r)}_{1,2;i,1} \qquad 1 \le i \le 2, \quad 2 \le r \le 3,$$

$$E^{(r)}_{1,3;i,1} \qquad 1 \le i \le 2, \quad 2 \le r \le 3,$$

and

$$E^{(r)}_{2,3;1,1} \qquad 1 \le r \le 4.$$

For the negative root elements, the allowed elements are

$$F^{(r)}_{1,2;1,j} \qquad 1 \le j \le 2, \quad 2 \le r \le 3,$$

$$F^{(r)}_{1,3;1,j} \qquad 1 \le j \le 2, \quad 2 \le r \le 3,$$

and

$$F^{(r)}_{2,3;1,1} \qquad 1 \le r \le 4.$$

Thus this example uses the parabolic presentation in an essential way: the first diagonal block has size 2*, and the elements* $E_{1,2}$*,* $E_{1,3}$*,* $F_{1,2}$*, and* $F_{1,3}$ *carry internal matrix indices. The basis is not the list above by itself, but the set of ordered monomials in these elements.*

7 Invariant elements in $U(\mathfrak{p})$

We now construct the elements of $U(\mathfrak{p})$ corresponding to the generators of the level ℓ shifted Yangian. Let π have N boxes, height n, level ℓ, and column heights $(q_1, \dots, q_\ell)$. Rows are numbered from top to bottom as before, and $n = \max(q_1, \dots, q_\ell)$.

7.1 Shifted matrix units and their normalization

The scalar shifts in [1, (9.1)] admit an auxiliary integer at least as large as the height of the pyramid. We denote this integer by κ to distinguish it from the fixed height n. We choose $\kappa = N$, which satisfies this bound, and set

$$\rho_r = \kappa - q_r - q_{r+1} - \cdots - q_\ell = N - q_r - q_{r+1} - \cdots - q_\ell, \qquad 1 \le r \le \ell.$$

For $1 \le h, k \le N$, define

$$\widetilde{e}_{hk} = (-1)^{\mathrm{col}(k)-\mathrm{col}(h)}\big(e_{hk} + \delta_{hk}\rho_{\mathrm{col}(h)}\big).$$

In the running example, this choice gives

$$(\rho_1, \dots, \rho_6) = (0, 1, 4, 7, 11, 13).$$

Here is an explicit comparison with the normalization obtained by taking the auxiliary integer to be n. Increasing that integer by a constant c is implemented by the algebra automorphism

$$\Phi_c : U(\mathfrak{g}) \longrightarrow U(\mathfrak{g}), \qquad \Phi_c(e_{hk}) = e_{hk} + c\delta_{hk}.$$

The scalar terms cancel in the matrix-unit commutator relations, so this is indeed an automorphism, with inverse Φ_{-c}. It preserves $U(\mathfrak{p})$ and fixes $\mathfrak{m}$ pointwise, since matrix units in $\mathfrak{m}$ are off-diagonal. Hence it preserves J_χ, commutes with pr_χ, and restricts to an automorphism of $W(\pi)$. Our choice is obtained from the height normalization by Φ_{N-n}. We keep n as the rank of the shifted Yangian throughout.

Assign the matrix units their **Kazhdan degrees**

$$\deg_K(e_{hk}) = \mathrm{col}(k) - \mathrm{col}(h) + 1.$$

This gives a filtration, rather than a Lie algebra grading: for homogeneous matrix units with nonzero commutator, its degree is one less than the sum of their degrees. On $U(\mathfrak{p})$ all generator degrees are positive. If $\mathrm{col}(h) \le \mathrm{col}(k)$, then $\widetilde{e}_{hk}$ lies in $F_{\deg_K(e_{hk})}U(\mathfrak{p})$.

7.2 The path matrix

For $1 \le i, j \le n$ and $r \ge 1$, define

$$\widetilde{T}_{ij}^{(r)} = \sum_{s=1}^{r} \sum_{\substack{i_1, j_1, \dots, i_s, j_s \\ \text{admissible}}} \widetilde{e}_{i_1 j_1} \cdots \widetilde{e}_{i_s j_s}.$$

All indices in the inner sum belong to $\{1, \dots, N\}$. A sequence is admissible when

$$\begin{gathered}
\sum_{t=1}^{s} \big(\mathrm{col}(j_t) - \mathrm{col}(i_t) + 1\big) = r, \\
\mathrm{col}(i_t) \le \mathrm{col}(j_t) \qquad (1 \le t \le s), \\
\mathrm{col}(j_t) < \mathrm{col}(i_{t+1}) \qquad (1 \le t < s), \\
\mathrm{row}(i_1) = i, \qquad \mathrm{row}(j_s) = j, \\
\mathrm{row}(j_t) = \mathrm{row}(i_{t+1}) \qquad (1 \le t < s).
\end{gathered}$$

The strict inequality between consecutive factors requires the path to move to a later column before starting its next factor. This is the all-positive-sign case $T_{i,j;0}^{(r)}$ of [1, (9.5)], with our choice of scalar shifts. Each summand belongs to $F_r U(\mathfrak{p})$. The inequalities also imply $\widetilde{T}_{ij}^{(r)} = 0$ for $r > \ell$, since the total degree counts the lengths of disjoint ordered column intervals.

Set $\widetilde{T}_{ij}^{(0)} = \delta_{ij}$ and collect the coefficients into

$$\widetilde{T}_{ij}(u) = \sum_{r \ge 0} \widetilde{T}_{ij}^{(r)} u^{-r}, \qquad \widetilde{T}(u) = \big(\widetilde{T}_{ij}(u)\big)_{1 \le i,j \le n}.$$

For example, the first two coefficients are

$$\widetilde{T}_{ij}^{(1)} = \sum_{\substack{\mathrm{row}(h)=i,\ \mathrm{row}(k)=j \\ \mathrm{col}(h)=\mathrm{col}(k)}} \widetilde{e}_{hk}$$

and

$$\widetilde{T}_{ij}^{(2)} = \sum_{\substack{\mathrm{row}(h)=i,\ \mathrm{row}(k)=j \\ \mathrm{col}(k)=\mathrm{col}(h)+1}} \widetilde{e}_{hk} + \sum_{\substack{\mathrm{row}(h_1)=i,\ \mathrm{row}(k_1)=\mathrm{row}(h_2),\ \mathrm{row}(k_2)=j \\ \mathrm{col}(h_1)=\mathrm{col}(k_1)<\mathrm{col}(h_2)=\mathrm{col}(k_2)}} \widetilde{e}_{h_1k_1}\widetilde{e}_{h_2k_2}.$$

Membership in $U(\mathfrak{p})$ alone does not imply twisted invariance. The invariant generators are selected from the Gauss factors using the mode bounds below.

7.3 Gauss factors and the Brundan–Kleshchev isomorphism

Let $\nu = (\nu_1, \dots, \nu_m)$ be admissible for σ. Since $\widetilde{T}(u)$ has constant term I_n, it has a unique formal block Gauss decomposition

$$\widetilde{T}(u) = \widehat{F}(u)\widehat{D}(u)\widehat{E}(u).$$

Here $\widehat{D}$ is block diagonal, and $\widehat{E}$ and $\widehat{F}$ are block upper and lower unitriangular, respectively. We use hats to distinguish the images of the Yangian generators from the inverse series $\widetilde{D}_a(u) = -D_a(u)^{-1}$ introduced earlier. Write

$$\begin{aligned}
\widehat{D}_a(u) &= I_{\nu_a} + \sum_{r>0} \big(\widehat{D}_{a;i,j}^{(r)}\big) u^{-r}, \\
\widehat{E}_a(u) &= \sum_{r>0} \big(\widehat{E}_{a;i,j}^{(r)}\big) u^{-r}, \\
\widehat{F}_a(u) &= \sum_{r>0} \big(\widehat{F}_{a;i,j}^{(r)}\big) u^{-r}.
\end{aligned}$$

The matrices have sizes $\nu_a \times \nu_a$, $\nu_a \times \nu_{a+1}$, and $\nu_{a+1} \times \nu_a$, respectively. All their coefficients lie in $U(\mathfrak{p})$.

Theorem 7.1 (Brundan–Kleshchev)**.** *For every admissible shape ν, there is a unique isomorphism*

$$Y_{n,\ell}(\sigma) \longrightarrow W(\pi)$$

specified on the parabolic generators, with all block and matrix indices in their defining ranges, by

$$\begin{aligned} D^{(r)}_{a;i,j} &\longmapsto \widehat{D}^{(r)}_{a;i,j} \qquad (r > 0), \\ E^{(r)}_{a;i,j} &\longmapsto \widehat{E}^{(r)}_{a;i,j} \qquad \big(r > s_{a,a+1}(\nu)\big), \\ F^{(r)}_{a;i,j} &\longmapsto \widehat{F}^{(r)}_{a;i,j} \qquad \big(r > s_{a+1,a}(\nu)\big). \end{aligned}$$

These images are twisted $\mathfrak{m}$-invariants and generate $W(\pi)$. The isomorphism is filtered for the canonical filtration on the shifted Yangian and the Kazhdan filtration on $W(\pi)$.

For the height normalization this is [1, Theorem 10.1], with the Gauss construction from Section 9 of that paper. The choice $\kappa = N$ follows by composing the isomorphism with Φ_{N-n}. The invariance argument and the proof of the isomorphism are given in Section 10. Only the coefficients in the stated mode ranges are asserted to be invariant; lower modes of the off-diagonal Gauss factors need not be invariant.

There is also a useful general projection argument. Let

$$U(\mathfrak{g})^{\mathfrak{m}} = \{z \in U(\mathfrak{g}) \mid [x, z] = 0 \text{ for all } x \in \mathfrak{m}\}.$$

Then

$$\mathrm{pr}_\chi\big(U(\mathfrak{g})^{\mathfrak{m}}\big) \subseteq W(\pi).$$

Indeed, $z - \mathrm{pr}_\chi(z) \in J_\chi$, and adjoint stability of J_χ gives $[x, \mathrm{pr}_\chi(z)] \in J_\chi$ for every $x \in \mathfrak{m}$.

The restriction of pr_χ to $U(\mathfrak{g})^{\mathfrak{m}}$ is multiplicative. If z, w centralize $\mathfrak{m}$, then $J_\chi w \subseteq J_\chi$, because

$$(x - \chi(x))w = w(x - \chi(x)) \in J_\chi.$$

Writing $z = a + b$ and $w = c + d$, with $a, c \in U(\mathfrak{p})$ and $b, d \in J_\chi$, gives

$$zw = ac + ad + bw, \qquad ad, bw \in J_\chi.$$

Therefore $\mathrm{pr}_\chi(zw) = ac = \mathrm{pr}_\chi(z)\,\mathrm{pr}_\chi(w)$. This supplies one way of obtaining twisted invariants by projecting ordinary centralizing elements.

Example 7.2 (A single column)**.** *Suppose π is a single column of height N. Then $n = N$, $\ell = 1$, $e_\pi = 0$, $\mathfrak{p} = \mathfrak{g}$, and $\mathfrak{m} = 0$. Consequently,*

$$W(\pi) = U(\mathfrak{gl}_N).$$

Since $q_1 = N$, our normalization gives $\rho_1 = N - q_1 = 0$ and $\widetilde{e}_{hk} = e_{hk}$. Every factor has Kazhdan degree 1. *All boxes have column number* 1, *so the condition* $\mathrm{col}(j_t) < \mathrm{col}(i_{t+1})$ *excludes paths with two or more factors. With $E = (e_{ij})_{1 \le i,j \le N}$, we obtain*

$$\widetilde{T}_{ij}^{(1)} = e_{ij}, \qquad \widetilde{T}_{ij}^{(r)} = 0 \quad (r > 1), \qquad \widetilde{T}(u) = I + Eu^{-1}.$$

Choose the admissible shape $\nu = (N)$, so there is one diagonal block and $\widehat{D}_1(u) = \widetilde{T}(u)$. The truncation is therefore respected:

$$D_{1;i,j}^{(1)} \longmapsto e_{ij}, \qquad D_{1;i,j}^{(r)} = 0 \quad (r > 1).$$

In this shape the RTT matrix equals $D_1(u)$, and the isomorphism becomes the usual level-one evaluation isomorphism

$$Y_{N,1}(0) \cong U(\mathfrak{gl}_N).$$

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